JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SUNDAY 05th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let α, β be the roots of the equation $x^2 - x + p = 0$ and γ, δ be the roots the equation $x^2 - 4x + q = 0$; $p, q \in Z$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then $|p+q|$ equals:

(1) 16 (2) 32

(3) 34 (4) 38

Sol. Let $\alpha = a, \beta = ar, \gamma = ar^2, \delta = ar^3$

$a + ar = 1$

$ar^2 + ar^3 = 4$

$\Rightarrow ar^2(1 + r) = 4$

$\Rightarrow r = 2, a = \frac{1}{3}$ (reject as $p \in z$)

or $r = -2, a = -1$

Now $|P+q| = |a(ar) + ar^2 - ar^3|$

$= |a^2r(1 + r^4)|$

$|1 \times -2 \times 17| = 34$

2. Let $z_1, z_2 \in C$ be the distinct solutions of the equation $z^2 + 4z - (1 + 12i) = 0$. Then $|z_1|^2 + |z_2|^2$ is equal to :

(1) 18 (2) 22

(3) 29 (4) 34

Sol. $Z^2 + 4Z - (1 + 12i) = 0$

$\Rightarrow Z = \frac{-4 \pm \sqrt{16 + 4(1+12i)}}{2}$

$\Rightarrow Z = -2 \pm \sqrt{5+12i}$

$\Rightarrow Z = -2 \pm (3 + 2i)$

$\Rightarrow Z = 1 + 2i, -5 - 2i$

$\therefore |Z_1|^2 + |Z_2|^2$

$= 5 + 29$

$= 34$

3. If $f : N \to Z$ is defined by

$$f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}, k \in N,$$ and

$\sum_{n=1}^{k} f(n) = 98$, then k is equal to:

(1) 3 (2) 4

(3) 5 (4) 6

Sol. $$f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}$$

$$\sum_{n=1}^{k} f(n) = \begin{vmatrix} \frac{k(k+1)}{2} & -1 & -5 \\ -2\frac{k(k+1)(2k+1)}{6} & 3(2k+1) & 2k+1 \\ -3\frac{k^2(k+1)^2}{4} & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix} = 98$$

$\Rightarrow \frac{k(k+1)(2k+1)}{2} \cdot \frac{7}{3} = 98$

$\Rightarrow k(k + 1)(2k + 1) = 84$

$\Rightarrow k = 3$

4. Let M be a 3×3 matrix such that

$M\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, M\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix}$ and

$M\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$. If $M\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$, then $x + y + z$ equals :

(1) 4 (2) 5

(3) 7 (4) 11

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Sol. Let $M = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

then $M\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix} \Rightarrow a_1 = 1, b_1 = 2, c_1 = 3$

$M\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix} \Rightarrow a_2 = 0, b_2 = 1, c_2 = 2$

& $M\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix} \Rightarrow a_3 = -1, b_3 = 1, c_3 = 1$

Now $M\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$

$\Rightarrow x + 0y - z = 1$

$2x + y + z = 7$

$3x + 2y + z = 11$

on solving $x = 2, y = 2, z = 1$

$\therefore \quad x + y + z = 5$

5. If the sum of the first 10 terms of the series $\frac{1}{1+1^4\times4}+\frac{2}{1+2^4\times4}+\frac{3}{1+3^4\times4}+\frac{4}{1+4^4\times4}+\ldots$ is $\frac{m}{n}$, gcd (m, n) = 1, then m + n is equal to :

(1) 256 (2) 264

(3) 276 (4) 284

Sol. $T_r = \frac{r}{1+4r^4} = \frac{r}{4r^4+4r^2+1-4r^2} = \frac{r}{(2r^2+1)^2-(2r)^2}$

$= \frac{r}{(2r^2+2r+1)(2r^2-2r+1)}$

$S_{10} = \frac{1}{4}\left(\frac{1}{1}-\frac{1}{221}\right) = \frac{55}{221} = \frac{m}{n}$

$\therefore m + n = 276$

6. Let $A_1, A_2, A_3, \ldots\ldots, A_{39}$ be 39 arithmetic means between the numbers 59 and 159. Then the mean of A_{25}, A_{28}, A_{31} and A_{36} is equal to:

(1) 129 (2) 136

(3) 131.50 (4) 134

Sol. 59 $A_1A_2A_3 \ldots A_{39}$ 159

$d = \frac{159-59}{39+1} = \frac{5}{2}$

Now, $A_{25} = 59 + 25d = 121.5$

$A_{28} = 59 + 28d = 129$

$A_{31} = 59 + 31d = 136.5$

$A_{36} = 59 + 36d = 149$

Mean of

$A_{25}, A_{28}, A_{31}, A_{36} = \frac{121.5+129+136.5+149}{4}$

$= \frac{536}{4} = 134$

7. The coefficient of x^2 in the expansion of $\left(2x^2+\frac{1}{x}\right)^{10}, x \neq 0$, is :

(1) 3240 (2) 3360

(3) 3480 (4) 3600

Sol. $T_{r+1} = {}^{10}C_r(2x^2)^{10-r}(1/x)^r$

$= {}^{10}C_r 2^{10-r} x^{20-2r-r}$

$= {}^{10}C_r 2^{10-r} x^{20-3r}$

$\Rightarrow 20 - 3r = 2$

$r = 6$

So required coefficient is ${}^{10}C_6 2^4$

8. The probabilities that players A and B of a team are selected for the captaincy for a tournament are 0.6 and 0.4, respectively. If A is selected the captain, the probability that the team wins the tournament is 0.8 and if B is selected the captain, the probability that the team wins the tournament is 0.7. Then the probability, that the team wins the tournament, is :

(1) 0.74 (2) 0.76

(3) 0.72 (4) 0.78

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Sol. $P(Win) = P(A \cap Win) + P(B \cap Win)$

$= P(A) \cdot P\left(\frac{Win}{A}\right) + P(B) \cdot P\left(\frac{Win}{B}\right)$

$= (0.6)(0.8) + (0.4)(0.7)$

$= 0.48 + 0.28 = 0.76$

9. A box contains 5 blue, 6 yellow and 4 red balls. The number of ways, of drawing 8 balls containing at least two balls of each colour, is:

(1) 4100 (2) 4140

(3) 4230 (4) 4290

Sol. (But in this questions not mentioned that balls are distinct/alike)?

Blue – 5 Yellow – 6 Red – 4

Reqd. No. of ways

$= {}^5C_2 \times {}^6C_2 \times {}^4C_4 + {}^5C_2 \times {}^6C_3 \times {}^4C_3 + {}^5C_2 \times {}^6C_4 \times {}^4C_2 + {}^5C_3 \times {}^6C_2 \times {}^4C_3 + {}^5C_3 \times {}^6C_3 \times {}^4C_2 + {}^5C_4 \times {}^6C_2 \times {}^4C_2$

$= 10 \times 15 \times 1 + 10 \times 20 \times 4 + 10 \times 15 \times 6 + 10 \times 15 \times 4 + 10 \times 20 \times 6 + 5 \times 15 \times 6$

$= 150 + 800 + 900 + 600 + 1200 + 450$

$= 4100$

10. A variable X takes values 0, 0, 2, 6, 12, 20,..... n(n–1) with frequencies ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ${}^nC_3$, ${}^nC_4$, ${}^nC_5$,.....${}^nC_n$, respectively. If the mean of this data is 60, then its median is:

(1) 56 (2) 42

(3) 72 (4) 90

Sol. $\bar{x} = \frac{0.{}^nC_0 + 0.{}^nC_1 + 2.{}^nC_2 + ... + n(n-1).nC_n}{{}^nC_0 + {}^nC_1 + {}^nC_2 + ... + nC_n}$

$= \frac{\sum_{r=0}^{n} r(r-1){}^nC_r}{2^n}$

$= \frac{\sum_{r=1}^{n} n(r-1)^{n-1}C_{r-1}}{2^n} = \frac{\sum_{r=2}^{n} n(n-1)^{n-2}C_{r-2}}{2^n}$

$= \frac{n(n-1)2^{n-2}}{2^n} = \frac{n(n-1)}{4}$

$\Rightarrow \frac{n(n-1)}{4} = 60 \Rightarrow n^2 - n - 240 = 0$

$\Rightarrow n = 16, -15$ (Reject)

$N = \sum f_i = {}^{16}C_0 + {}^{16}C_1 + ... + {}^{16}C_{16} = 2^{16}$

$N = 2^{16}$ (Even)

$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2}+1\right)^{th}}{2}$ term

$\frac{N}{2} = 2^{15} = 32768$

x_i	f_i
0	${}^{16}C_0$
0	${}^{16}C_1$
2	${}^{16}C_2$
6	${}^{16}C_3$
⋮	⋮
⋮	⋮
⋮	⋮
240	${}^{16}C_{16}$

x_i	f_i	CF
0	${}^{16}C_0$	1
0	${}^{16}C_1$	16
2	${}^{16}C_2$	137
6	${}^{16}C_3$	697
12	${}^{16}C_4$	2517
20	${}^{16}C_5$	6885
30	${}^{16}C_6$	14893
42	${}^{16}C_7$	26333
56	${}^{16}C_8$	39203 $\rightarrow \frac{N}{2}$ & $\left(\frac{N}{2}+1\right)$ lies
72	${}^{16}C_9$	$\frac{n(n-1)2^{n-2}}{2^n}$
⋮	⋮	⋮
⋮	⋮	⋮
⋮	⋮	⋮

$\Rightarrow \text{median} = \frac{56+56}{2} = 56$

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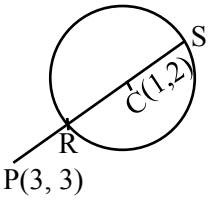
11. Let the point P be the vertex of the parabola $y = x^2 - 6x + 12$. If a line passing through the point P intersects the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ at the points R and S, then the maximum value of $(PR + PS)^2$ is:

(1) 10 (2) 20

(3) 25 (4) 5

Sol. Parabola $y = x^2 - 6x + 12 = (x-3)^2 + 3$

$y-3 = (x-3)^2 \Rightarrow$ Vertex P (3, 3)



$x^2 + y^2 - 2x - 4y + 3 = 0$

$C(1, 2); r = \sqrt{2}$

For max. of $(PR + PS)^2$

RS should be diameter

$(PR + PS)^2_{max} = (PC - r + PC + r)^2$

$= (2PC)^2 = 4(PC)^2$

$= 4 \times 5 = 20$

12. Let the directrix of the parabola P : $y^2 = 8x$, cut x-axis at the point A. Let B (α, β), $\alpha > 1$, be a point on P such that the slope of AB is 3/5. If BC is a focal chord of P, then six times the area of ΔABC is:

(1) 80 (2) 160

(3) 174 (4) 192

Sol. Let $B(2t^2, 4t) \Rightarrow C\left(\frac{2}{t^2}, \frac{-4}{t}\right); A(-2, 0)$

given $m_{AB} = \frac{4t}{2t^2+2} = \frac{3}{5}$

$20t = 6t^2 + 6$

$3t^2 - 10t + 3 = 0$

$t = 3, \frac{1}{3}$ but $t > 1$ (given)

$\Rightarrow P(18,12)$, $Q\left(\frac{2}{9}, \frac{-4}{3}\right)$

$$A = \left|\frac{1}{2}\begin{vmatrix} -2 & 0 & 1 \\ 18 & 12 & 1 \\ \frac{2}{9} & \frac{-4}{3} & 1 \end{vmatrix}\right| = \frac{80}{3}$$

$\Rightarrow 6A = 160$

13. Let the eccentricity e of a hyperbola satisfy the equation $6e^2 - 11e + 3 = 0$. If the foci of the hyperbola are (3, 5) (3, –4), then the length of its latus rectum is:

(1) $\frac{11}{3}$ (2) $\frac{17}{3}$

(3) $\frac{15}{2}$ (4) $\frac{17}{2}$

Sol. $S_1(3,5)$ & $S_2(3, -4)$

$\because S_1S_2 = 2ae$

$\Rightarrow 9 = 2ae$ ...(1)

and $\because 6e^2 - 11e + 3 = 0$

$\Rightarrow e = \frac{3}{2}$, $e = \frac{1}{3} < 1$ (rejected)

From (1); a = 3

$\Rightarrow LR = 2 \cdot \frac{b^2}{a} = 2a(e^2 - 1) = 2.3\left(\frac{9}{4} - 1\right) = \frac{15}{2}$

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14. Let a triangle PQR be such that P and Q lie on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ and are at a distance of 6 units from R (1, 2, 3). If (α, β, γ) is the centroid of ΔPQR, then $\alpha+\beta+\gamma$ is equal to:

(1) 4 (2) 5

(3) 6 (4) 8

Sol. Let the point P is $(8\lambda-3, 2\lambda+4, 2\lambda-1)$

$PR = 6 \Rightarrow PR^2 = 36$

$(8\lambda-4)^2 + (2\lambda+2)^2 + (2\lambda-4)^2 = 36$

$(4\lambda-2)^2 + (\lambda+1)^2 + (\lambda-2)^2 = 9$

$\lambda^2 - \lambda = 0 \Rightarrow \lambda = 0, 1$

$\lambda = 0 \Rightarrow P(-3, 4, -1)$

$\lambda = 1 \Rightarrow P(5, 6, 1)$

R(1, 2, 3)

G(1, 4, 1)

$= \alpha+\beta+\gamma = 6$

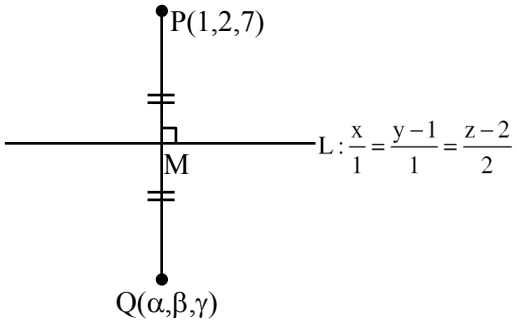
15. If the distance of the point (a, 2, 5) from the image of the point (1, 2, 7) in the line $\frac{x}{1}=\frac{y-1}{1}=\frac{z-2}{2}$ is 4, then the sum of all possible values of a is equal to :

(1) 11 (2) 9

(3) 6 (4) 4

Sol. $PQ \perp L \Rightarrow (\alpha-1) + (\beta-2) + 2(\gamma-7) = 0$

$\Rightarrow \alpha+\beta+2\gamma = 17 \quad \ldots(1)$



M is mid point of PQ which will satisfy L

$\frac{\alpha+1}{2}=\frac{\frac{\beta+2}{2}-1}{1}=\frac{\frac{\gamma+7}{2}-2}{2}$

$\Rightarrow \frac{\alpha+1}{2}=\frac{\beta}{2}=\frac{\gamma+3}{4}$

$\Rightarrow \alpha+1 = \beta \quad \ldots(2)$

and $2\beta = \gamma+3 \quad \ldots(3)$

$\Rightarrow \alpha = 3, \beta = 4, \gamma = 5$

Distance from (a, 2, 5) is $= \sqrt{(a-3)^2+4+0} = 4$

$(a-3)^2 + 4 = 16 \Rightarrow a^2 - 6a - 3 = 0 \Rightarrow$ sum of values of a = 6

16. Let O be the origin, $\overrightarrow{OP}=\vec{a}$ and $\overrightarrow{OQ}=\vec{b}$. If R is the point on $\overrightarrow{OP}$ such that $\overrightarrow{OP}=5\overrightarrow{OR}$, and M is the point such that $\overrightarrow{OQ}=5\overrightarrow{RM}$, then $\overrightarrow{PM}$ is equal to :

(1) $\frac{1}{5}(\vec{a}-4\vec{b})$ (2) $\frac{1}{5}(\vec{b}-4\vec{a})$

(3) $\frac{1}{5}(-\vec{a}+4\vec{b})$ (4) $\frac{1}{5}(-\vec{b}+4\vec{a})$

Sol. $\overrightarrow{OR}=\frac{\overrightarrow{OP}}{5}=\frac{\vec{a}}{5}$

$\overrightarrow{RM}=\frac{\overrightarrow{OQ}}{5}=\frac{\vec{b}}{5}$

$\overrightarrow{OM}-\overrightarrow{OR}=\frac{\vec{b}}{5}$

$\overrightarrow{OM}=\frac{\vec{a}+\vec{b}}{5}$

$\overrightarrow{PM}=\overrightarrow{OM}-\overrightarrow{OP}$

$\frac{\vec{a}+\vec{b}}{5}-\vec{a}$

$=\frac{\vec{b}-4\vec{a}}{5}$

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17. Let $f(x) = \lim\limits_{y\to 0}\frac{(1-\cos(xy))\tan(xy)}{y^3}$. Then the number of solutions of the equation $f(x) = \sin x$, $x \in R$ is :

(1) 0 (2) 2

(3) 3 (4) 1

Sol. $f(x) = \lim\limits_{y\to 0}\frac{1-\cos(xy)}{(xy)^2}\cdot\frac{\tan xy}{xy}\cdot\frac{(xy)^3}{y^3}$

$f(x) = \frac{x^3}{2}$

$f(x) = \sin x$

$\frac{x^3}{2} = \sin x$

3 points of intersection

18. Let $(2^{1-a} + 2^{1+a})$, $f(a)$, $(3^a + 3^{-a})$ be a A. P. and α be the minimum value of $f(a)$. Then the value of the integral $\int\limits_{\log_e(\alpha-1)}^{\log_e(\alpha)}\frac{dx}{(e^{2x}-e^{-2x})}$ is :

(1) $\frac{1}{2}\log_e\left(\frac{4}{3}\right)$ (2) $\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

(3) $\frac{1}{2}\log_e\left(\frac{8}{5}\right)$ (4) $\frac{1}{4}\log_e\left(\frac{8}{5}\right)$

Sol. $f(a) = \frac{1}{2}(2(2^a + 2^{-a}) + (3^a + 3^{-a}))$

$a = \frac{1}{2}(2\times 2 + 2) = 3$

$I = \int\limits_{\ln(3-1)}^{\ln 3}\frac{e^{2x}}{e^{4x}-1}dx$

put $e^{2x} = t$

$I = \frac{1}{2}\int\limits_4^9\frac{dt}{t^2-1}$

$I = \frac{1}{4}\left(\ln\left(\frac{t-1}{t+1}\right)\right)_4^9$

$I = \frac{1}{4}\ln\left(\frac{4}{3}\right)$

19. Let $f : [1, \infty) \to R$ be a differentiable function defined as $f(x) = \int\limits_1^x f(t)dt + (1-x)(\log_e x - 1) + e$.

Then the value fo f(f(1)) is :

(1) $(1 + e^e)$ (2) $(1 + e)$

(3) $(1 + e + e^e)$ (4) $1 + 2e$

Sol. $f(1) = \int\limits_1^1 f(t)dt + (1-1)(\ln 1 - 1) + e$

$f(1) = e$

$f(x) = \int\limits_1^x f(t)dt + (1-x)(\ln x - 1) + e$

diff. w.r.t. x

$f'(x) = f(x) + \frac{1}{x} - \ln x$

I.F. $= e^{\int -1dx} = e^{-x}$

$f(x)\, e^{-x} = \int e^{-x}\left(\frac{1}{x} - \ell nx\right)dx$

$f(x)\, e^{-x} = e^{-x}\ln x + C$

put $x = 1$, $y = e$

$\Rightarrow C = 1$

$f(x)e^{-x} = e^{-x}|nx + 1$

$f(x) = e^x + \ln x$

$f(e) = e^e + 1$

20. Let f(x) and g(x) be twice differentiable functions satisfying f '' (x) = g''(x) for all $x \in R$, f '(1) = 2g'(1) = 4 and g(2) = 3f(2) = 9. Then f(25) – g(25) is equal to :

(1) 20 (2) 40

(3) –20 (4) –40

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Sol. $f''(x) = g''(x)$

Integrate

$f'(x) = g'(x) + C_1$

Put $x = 1$

$f'(1) = g'(1) + C_1$

$4 = 2 + C_1$

$C_1 = 2$

$\Rightarrow f'(x) = g'(x) + 2$

$f(x) = g(x) + 2x + C_2$

Put $x = 2$

$f(2) = g(2) + 4 + C_2$

$3 = 9 + 4 + C_2$

$C_2 = -10$

$f(x) = g(x) + 2x - 10$

Put $x = 25$

$f(x) - g(25) = 40$

SECTION-B

21. Let A = {1, 4, 7} and B = {2, 3, 8}. Then the number of elements, in the relation $R = \{((a_1, b_1), (a_2, b_2)) \in ((A \times B) \times (A \times B)) : a_1 + b_2$ divides $a_2 + b_1\}$ is ……..

Sol.

A / B	2	3	8
1	3	4	9
4	6	7	12
7	9	10	15

These are possible sum of $a_i + b_j$.

Now $b_1 + a_2$ divides $a_1 + b_2$

So,

$a_1 + b_2$	$a_2 + b_1$	No. of pairs
15	3,15	2
12	3,4,6,12	4
10	10	1
9	3,9	6
7	7	1
6	3,6	2
4	4	1
3	3	1
		18

Number of relations = 18

22. From the point (–1, –1), two rays are sent making angles of 45° with the line x + y = 0. These rays get reflected from the mirror x + 2y = 1. If the equations of the reflected rays are ax + by = 9 and cx + dy = 7, a, b, c, d ∈ Z, then the value of ad + bc is ……..

Sol. $\tan 45° = \left|\frac{m+1}{1-m}\right|$

$\Rightarrow \pm 1 = \frac{m+1}{1-m}$

$\Rightarrow m = 0$ & one line is perpendicular to x axis.

$L_1 : x = -1, L_2 : y = -1$

Now taking mirror image of x + 1 = 0 in x + 2y = 1

$A' : \frac{x+1}{1} = \frac{y+1}{2} = \frac{-2(-1-2-1)}{1^2+2^2}$

$x = \frac{3}{5}, y = \frac{11}{5}$ $A'\left(\frac{3}{5}, \frac{11}{5}\right)$

line = 3x – 4y + 7 = 0

Now taking mirror image of y + 1 = 0 in x + 2y = 1

$A'' : \frac{x+1}{1} = \frac{y+1}{2} = \frac{-2(-1-2-1)}{1^2+2^2}$

$x = \frac{3}{5}, y = \frac{11}{5}$

$A''\left(\frac{3}{5}, \frac{11}{5}\right)$

$\text{line} : (y+1) = \frac{\left(\frac{11}{5}+1\right)}{\left(\frac{3}{5}-3\right)}(x-3)$

4x + 3y = 9

comparing 4x + 3y = 9 with ax + by = 9 &

a = 4, b = 3

by line

– 3x + 4y = 7

comparing with

cx + dy = 7

∴ c = –3, d = 4

So, ad + bc

= 4 × 4 + 3 × –3

= 16 – 9 = 7

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23. If $S = \left\{\theta \in [-\pi, \pi] : \cos\theta\cos\frac{5\theta}{2} = \cos 7\theta\cos\frac{7\theta}{2}\right\}$, then n(S) is equal to .....

Sol. $\cos\theta\cos\frac{5\theta}{2} = \cos 7\theta\cos\frac{7\theta}{2}$

$\cos\frac{7\theta}{2} + \cos\frac{3\theta}{2} = \cos\frac{21\theta}{2} + \cos\frac{7\theta}{2}$

$\cos\frac{21\theta}{2} - \cos\frac{3\theta}{2} = 0$

$-2\sin 6\theta\sin\frac{9\theta}{2} = 0$

$\sin 6\theta = 0$

$\theta = 0, \pm\frac{\pi}{6}, \pm\frac{2\pi}{6}, \ldots \pm\frac{5\pi}{6}, \pm\pi$ (13 solutions)

$\sin\frac{9\theta}{2} = 0$

$\theta = \pm\frac{2\pi}{9}, \pm\frac{4\pi}{9}, \pm\frac{8\pi}{9}$ (6 more solutions)

Total = 19 solutions

24. Let $f : R \to R$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$, $x \in R$. Let maximum value of f on R be α. If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3$, $\beta > 0$, is α^2, then $30\beta^3$ is equal to .....

Sol. $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$ ...(1)

Put $x \to \frac{\pi}{2} - x$

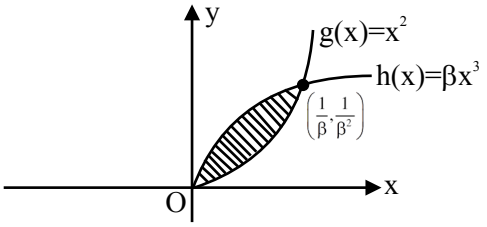
$\Rightarrow f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x$ ...(2)

From (1) & (2)

$f(x) = \frac{1}{8}(3\cos x - \sin x)$

$f_{max} = \frac{\sqrt{10}}{8}$

y = g(x) & y = h(x) intersect as shown in the figure



$\therefore$ Area bounded $= \Delta = \left|\int_0^{\frac{1}{\beta}} (\beta x^3 - x^2)\, dx\right|$

$= \frac{1}{12\beta^3} = \alpha^2$ (given)

$\Rightarrow 30\beta^3 = 16$

25. Let y = y(x) be the solution of the differential equation $(\tan x)^{1/2}dy = (\sec^3 x - (\tan x)^{3/2}y)dx$, $0 < x < \frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then α^4 equals .....

Sol. $\frac{dy}{dx} + y\tan x = \frac{\sec^3 x}{\sqrt{\tan x}}$

IF $= e^{\int \tan x dx} = \sec x$

$y\sec x = \int \frac{\sec^4 x}{\sqrt{\tan x}} dx$

$y\sec x = \int \frac{(1 + \tan^2 x)}{\sqrt{\tan x}} \sec^2 x dx$

$\tan x = t$

$y\sec x = 2\sqrt{\tan x} + \frac{2}{5}(\tan x)^{5/2} + c$

$y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$

$\frac{6\sqrt{2}}{5} \times \sqrt{2} = 2 + \frac{2}{5} + c \Rightarrow \frac{12}{5} = \frac{12}{5} + c$

$\Rightarrow c = 0$

$y(\pi/3) = \frac{8}{5}.3^{1/4} \Rightarrow \alpha = 2 \times 3^{\frac{1}{4}}$

$\therefore \alpha^4 = 48$

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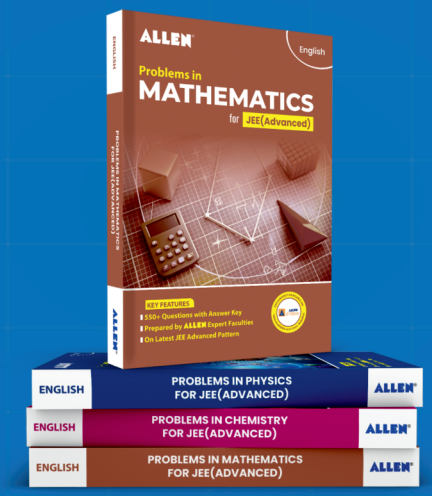


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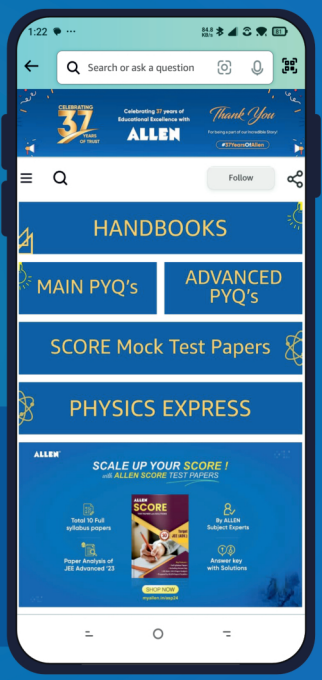
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